MarketMind — AI Financial Intelligence Platform Comprehensive Project Documentation

Introduction

MarketMind is an AI-powered financial intelligence platform designed to help Indian retail investors make better, faster, and more informed investment decisions. Modern financial markets generate an enormous volume of information every day, including news articles, corporate filings, stock price movements, regulatory disclosures, and macroeconomic updates. Retail investors often struggle to process this information effectively and identify which signals are actually relevant to their portfolios.
MarketMind solves this challenge by combining real-time market data, AI-powered analysis, and portfolio personalization. The platform acts as an intelligent layer between financial data sources and the investor, automatically filtering, analyzing, and presenting insights that matter most to each user. Instead of manually reading dozens of news articles or scanning complex financial reports, investors receive clear, structured insights generated by AI.
The platform was designed for the ET AI Hackathon by Blogy Hackathon 2026 and merges two challenge themes: AI for the Indian Investor and AI Native News Experiences. By combining these two ideas, MarketMind demonstrates how artificial intelligence can transform financial research into an interactive and personalized experience.

Problem Statement

Retail investors face several major challenges when navigating financial markets.
The first challenge is information overload. Every day thousands of financial articles, corporate filings, and regulatory updates are published. Identifying which pieces of information actually matter for a specific investment portfolio is extremely difficult.
The second challenge is lack of personalization. Most financial platforms provide generic recommendations that are not tailored to a user's holdings, risk profile, or financial goals. As a result, investors often receive insights that are not relevant to their investment strategy.
The third challenge is delayed discovery of opportunities. Important signals such as insider trades, bulk deals, or technical breakouts may occur long before they are widely discussed in the news. Retail investors often discover these opportunities too late.
MarketMind addresses these problems by using artificial intelligence to continuously monitor financial data sources and surface actionable insights in real time.

Solution Overview

MarketMind provides a unified dashboard where investors can track their portfolio, read AI summarized financial news, analyze technical patterns, and receive market signals. The platform operates using a multi agent AI architecture in which specialized AI components perform different financial analysis tasks.
These agents include a Signal Agent that detects insider trading activity and bulk deals, a Pattern Agent that analyzes chart patterns and price behavior, a News Agent that summarizes financial news articles and tags them with sentiment, and a Chat Agent that allows users to ask portfolio specific questions in natural language.
All agents operate using a central orchestration layer that routes requests to the correct analysis component. The orchestrator also enriches every AI request with contextual information such as the user's portfolio holdings, current market prices, and recent financial news. This ensures that the

insights generated by the AI system are highly personalized and relevant.
The result is a system that acts as a financial intelligence assistant rather than just a market data dashboard.

System Architecture

The architecture of MarketMind consists of five primary layers: the User Layer, the Frontend Interface, the AI Orchestration Layer, the AI Agent Layer, and the Data Layer.
The User Layer represents the investor interacting with the application. Users access the system through a web dashboard where they can import their portfolio, view signals, read news summaries, and interact with the AI assistant.
The Frontend Interface provides the visual dashboard. This layer displays portfolio analytics, signal alerts, financial news summaries, and AI chat responses. It also handles user inputs such as portfolio updates and investment queries.
The AI Orchestration Layer acts as the central brain of the system. When a user performs an action, the orchestrator determines which AI agent should handle the request. It also constructs the contextual prompt that will be sent to the AI model, ensuring that the model receives relevant portfolio data and financial context.
The AI Agent Layer contains specialized agents that perform different financial intelligence tasks. Each agent is optimized for a particular type of analysis. For example, the Signal Agent analyzes regulatory disclosures and trading activity, while the News Agent processes financial articles and generates structured summaries.
Finally, the Data Layer aggregates information from multiple sources such as stock market APIs, financial news feeds, and regulatory disclosures. These data sources provide the raw inputs used by the AI agents to generate insights.

AI Agent System

The multi agent design is one of the core innovations of MarketMind. Instead of relying on a single AI model to perform all tasks, the platform uses several specialized agents that collaborate to produce high quality financial insights.
The Signal Agent continuously monitors market activity for signals such as insider trades, bulk deals, and unusual price movements. These signals are analyzed using historical patterns to determine their significance.
The Pattern Agent analyzes stock price data to detect technical chart patterns such as breakouts, double bottoms, and ascending triangles. These patterns are commonly used by traders to identify potential price movements.
The News Agent processes financial news articles from sources such as Economic Times. The agent summarizes long articles into concise bullet points and assigns sentiment labels indicating whether the news is bullish, bearish, or neutral.
The Chat Agent enables investors to interact with the system through natural language queries. By incorporating portfolio data into each response, the agent can provide personalized recommendations and scenario analysis.
The Finance Agent handles personal financial planning features such as retirement projections, tax optimization, and financial health scoring.

Technology Stack

MarketMind uses a modern technology stack designed for scalability and rapid development.

The frontend is built using HTML, CSS, and JavaScript, providing a lightweight but interactive user interface. Data visualization libraries such as Chart.js are used to display portfolio analytics and market trends.
The backend is built using Node.js and Express, which provide a simple and efficient framework for building API services. The backend handles requests from the frontend and coordinates interactions with the AI services and data sources.
Artificial intelligence capabilities are powered by Google Gemini models accessed through API integration. These models are used for reasoning tasks such as summarization, pattern explanation, and financial analysis.
Market data is obtained from multiple sources including stock exchange feeds, regulatory filings, and financial news APIs. These data sources ensure that the insights generated by the platform are based on real market information.

Impact and Future Potential

MarketMind has the potential to significantly improve the investment experience for retail investors. By automating financial research and surfacing actionable insights, the platform reduces the time required to analyze market information from hours to seconds.
The system also democratizes access to sophisticated financial analysis tools that were previously available only to professional traders and institutional investors. Features such as signal detection, portfolio aware AI chat, and retirement planning tools help users make better financial decisions.
Future development could expand the platform to include additional capabilities such as earnings call analysis, options strategy recommendations, voice based financial assistants, and multilingual financial news translation.
As artificial intelligence continues to transform financial technology, platforms like MarketMind represent the next generation of intelligent investment tools.

Market Opportunity

The Indian retail investing ecosystem is expanding at rocket pace, like a financial supernova that keeps pulling new investors into orbit. Platforms that simplify financial intelligence stand in a prime position to capture this surge.
1. Rapid Growth of Retail Investors
India has seen a massive increase in individual investors over the last few years. According to the National Stock Exchange of India, the number of registered investors in the country has crossed 150 million, driven by digital trading platforms and rising financial awareness. Millions of first-time investors are entering the market every year, creating a huge demand for tools that simplify investment research.
2. Explosion of Financial Content
Financial information is growing at an overwhelming rate. News platforms like The Economic Times and Moneycontrol publish thousands of articles, updates, and reports every month. Retail investors struggle to identify which insights actually matter for their portfolio, creating a strong need for AI systems that filter and personalize financial information.

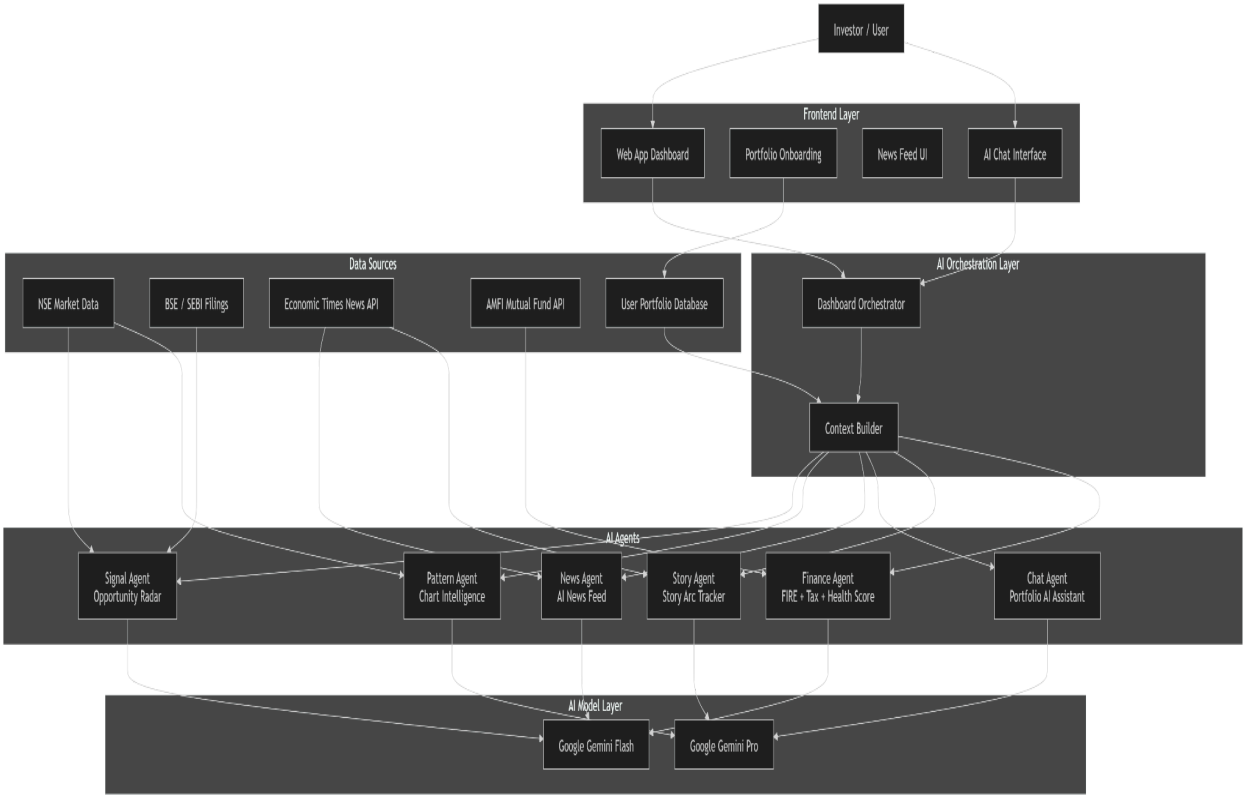
3. Shift Toward AI-Powered Finance
The global AI in fintech market is projected to grow rapidly over the next decade as financial institutions adopt automation and predictive analytics. Retail investors increasingly expect AI assistants that can analyze markets, summarize news, and generate personalized insights, similar to how AI copilots assist developers and creators.
4. Untapped Personalization Gap
While many investment apps provide market data and trading capabilities, very few platforms combine portfolio-aware AI, market signal detection, and financial planning in a single experience. MarketMind fills this gap by acting as an AI intelligence layer for retail investors, turning complex market signals into clear, actionable insights.

Architecture Document

Architecture Overview:-

The system is structured in five primary layers: User Layer, Frontend Layer, AI Orchestration Layer, AI Agent Layer, and Data Layer. Users interact with the system through a web dashboard where they can import their portfolio, view curated signals, read AI-summarized news, and ask financial questions through an AI chat interface. The frontend communicates with a dashboard orchestrator which determines which specialized agent should process each request. The Context Builder enriches every AI request by merging portfolio data with relevant market signals and news events. This ensures that the AI produces personalized responses tailored to the user's investments. Each AI agent performs a focused task: the Signal Agent identifies trading signals such as insider buys or bulk deals; the Pattern Agent detects technical chart patterns; the News Agent summarizes financial news and extracts sentiment; the Chat Agent powers portfolio-aware conversations; the Story Agent builds narrative timelines for major market events; and the Finance Agent handles retirement planning, tax optimization, and financial health scoring. The agents use Google Gemini models for reasoning and analysis. Gemini Flash is used for faster tasks like summarization and calculations, while Gemini Pro handles deeper reasoning such as portfolio analysis and financial storytelling. Market data is sourced from NSE market feeds, SEBI filings, Economic Times news APIs, AMFI mutual fund data, and the user’s portfolio database. The modular multi-agent architecture allows the platform to scale easily by adding additional agents for new financial capabilities

Tech Diagram :-


Investor / User
Frontend Layer
Web App Dashboard
Portfolio Onboarding
News Feed UI
AI Chat Interface
Data Sources
NSE Market Data
BSE / SEBI Filings
Economic Times News API
AMFI Mutual Fund API
User Portfolio Database
AI Orchestration Layer
Dashboard Orchestrator
Context Builder
AI Agents
Signal Agent
Opportunity Radar
Pattern Agent
Chart Intelligence
News Agent
AI News Feed
Story Agent
Story Arc Tracker
Finance Agent
FIRE + Tax + Health Score
Chat Agent
Portfolio AI Assistant
AI Model Layer
Google Gemini Flash
Google Gemini Pro